Auditing a clinical prediction model across patient subgroups with isitfair

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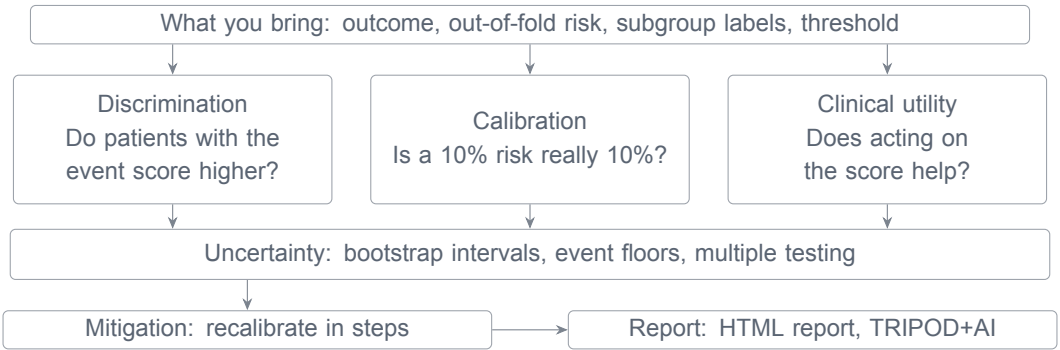
Inselspital, Bern University Hospital, and University of Bern

The worked example comes from a paper accepted at JAMIA in 2026

The talk has nine parts

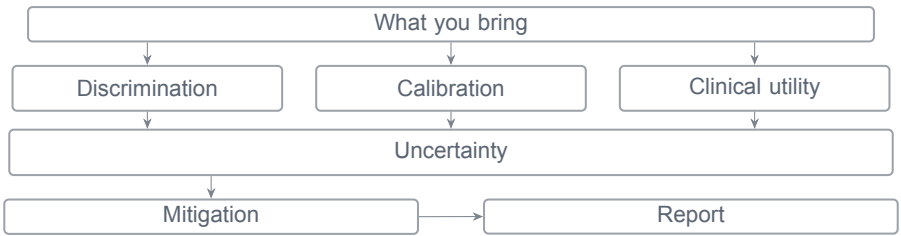
1. Why audit subgroups
2. Three axes
3. Inputs
4. Discrimination
5. Calibration
6. Clinical utility
7. Uncertainty
8. Mitigation
9. Reporting

The whole audit on one page



Read from top to bottom. Each part of the talk starts with this map, with its step marked.

Why audit subgroups



The model ranks patients well, for women and for men

The worked example: a model that estimates the risk of infection after surgery, at the end of the operation. 10,719 operations at Bern University Hospital, 2014–2022, with 504 infections.

AUROC 0.876 (95% CI 0.859–0.892)

AUROC is the area under the ROC curve. Women 0.878, men 0.875. Difference +0.004 (−0.030 to +0.038), p = 0.82.

By this measure the model looks ready to use.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Table 2 and Figure 2.

The risks the model prints are too high

Compare the printed risk with what happened to those patients:

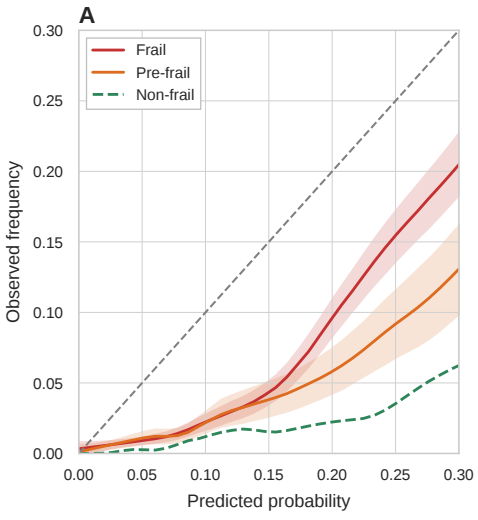
The model shows	5%	10%	20%	30%
Observed infection rate	0.9%	2.1%	8.3%	18.3%

Observed rates come from the smoothed calibration curve of all 10,719 operations. On average, the printed risk is 6.2 percentage points away from the observed rate.

Part of this error comes from the data. Before 2018, fewer infections were recorded: 4.70% of operations, against 8.28% from 2018 onward.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Overall calibration curve (subgroup curves in Figure 1A); expected calibration error in Table 3.

The size of the error depends on physiological reserve



Physiological reserve: how well a patient can tolerate a complication. It is built from the hospital record.

When the model shows 20%, the observed rate is:

Frail	9.6%
Pre-frail	5.8%
Non-frail	2.2%

The non-frail subgroup has only 6 infections. Its curve is dashed and has no interval.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Figure 1A.

The usefulness of the score differed by physiological reserve

Net benefit at a 10% threshold, per 1,000 operations, 2014–2022:

Frail	+17.21 (+10.61 to +23.73)
Pre-frail	−10.59 (−15.77 to −5.37)

A negative value means that acting on the score did more harm than acting on no one.

The pre-frail result did not hold in later data: +3.48 (−5.80 to +13.55) in 2018–2022, and +14.72 (−3.49 to +34.00) in a separate test on 2021–2022.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Table 4, Table 5 and Supplementary Table 2.

This tutorial follows the order of a real audit

1. What you must bring, and what makes your inputs unusable.
2. Three axes: what each measures, how to estimate it, how to read it.
3. Uncertainty: what an interval means, and when to withhold it.
4. Mitigation, in the order to try it.
5. A report you can give to a reviewer.

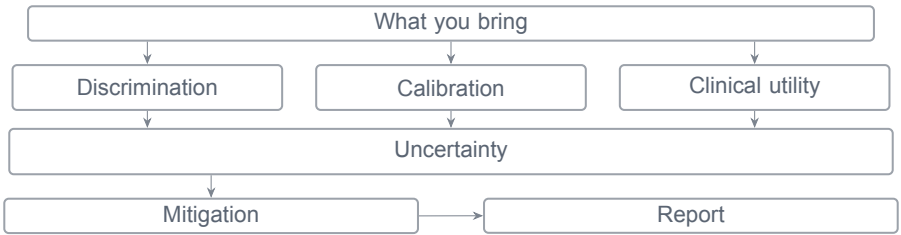
Slides with a formula also say it **in words**. **Common mistake** marks the error a slide helps you avoid.

All code on these slides runs on isitfair 0.4.

Six terms used throughout the talk

- **Rank and value.** A score can put patients in the right order and still print wrong risk values.
- **Threshold t.** The risk above which you act. Results at a threshold describe the decision rule. Another threshold gives other results.
- **Prevalence.** The share of a subgroup with the outcome. Some metrics change with prevalence alone.
- **Log-odds.** $\log(p/(1-p))$. It maps risks between 0 and 1 onto all real numbers, so a constant shift becomes one number.
- **Bootstrap interval.** Resample the patients 2,000 times and recompute. The middle 95% of the results is the interval.
- **Not estimable.** Too few events for an interval. The point estimate is still shown.

Three axes


What you bring
Discrimination
Calibration
Clinical utility
Uncertainty
Mitigation
Report

Three questions about one model

Discrimination

Do patients with the event get higher scores?

AUROC, AUPRC, sensitivity

Calibration

Is a 10% risk really 10%?

curve, intercept, slope

Clinical utility

Does acting on the score do more good than harm?

net benefit

A model can rank patients well and print wrong risks. A model can print correct risks and still help no one at any useful threshold.

Common mistake. Reporting only discrimination, because it is the familiar metric.

Every result has a subgroup value and a gap

For each axis you get two kinds of number:

- **Subgroup value**: the AUROC, calibration intercept or net benefit inside one subgroup.
- **Gap**: the difference between subgroups on the same metric.

A gap of zero can hide a model that fails equally in both subgroups. So isitfair always returns both: one row per subgroup, and a `GAP` row.

Common mistake. A fairness dashboard with gap bars and no subgroup performance.

How the score is used decides the primary axis

The score is shown as a probability
A risk on screen, or a threshold that triggers an action.

- Primary: calibration and clinical utility.
- Threshold gaps are descriptive.

The score is used as a ranking
A fixed number of patients reviewed, highest risk first.

- Primary: discrimination and error-rate gaps.
- Calibration matters less.

Decide this before you look at results, and write it down.

In the worked example, the score is a probability that triggers closer monitoring. So calibration and clinical utility were primary.

Choose subgroup axes before you see any results

- Prespecify them from clinical knowledge and the planned use.
- Use only attributes known when the prediction is made.
- Include clinical axes, such as physiological reserve, beside sex and age.
- Report case-mix axes, such as procedure cluster, separately. They describe the population.

Common mistake. Auditing only sex and age. In the worked example, sex showed no problem. Physiological reserve did.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods; Tables 2 and 3.

Physiological reserve in the worked example

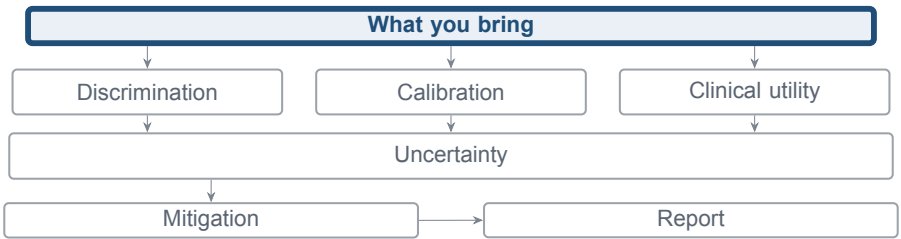
Label	Any of these	Operations	Infections
Frail	ASA ≥ 4, HFRS > 15, mFI-4 ≥ 2	6,139	400
Pre-frail	ASA 3, HFRS 5–15, mFI-4 = 1	3,558	98
Non-frail	none of these	1,022	6

ASA: American Society of Anesthesiologists physical status. HFRS: Hospital Frailty Risk Score. mFI-4: a four-item modified frailty index.

The labels describe a proxy built from the hospital record. They are not diagnoses of frailty.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods and Table 1.

Inputs


What you bring
Discrimination
Calibration
Clinical utility
Uncertainty
Mitigation
Report

Bring four inputs

1. `y_true`: the outcome, 0 or 1, one per patient.
2. `y_proba`: the predicted probability, for patients the model did not train on.
3. `sensitive_features`: a dict of subgroup labels, kept apart from the model's features.
4. `threshold`: one threshold *t*, fixed in advance.

isitfair never sees the model or its features. It evaluates predictions.

Notation used in the formulas

p_i is the predicted risk of patient i, and $Y_i \in \{0, 1\}$ the outcome.

For subgroup g: I_g its patients, $n_g = |I_g|$ their number, n_g^+ its events, and $\pi_g^+ = n_g^+ / n_g$ its prevalence.

A patient is flagged when $p_i \geq t$. This gives the counts $TP_g(t)$, $FP_g(t)$, $FN_g(t)$ and $TN_g(t)$.

An event is an outcome that happened. In the worked example, an event is an infection.

Use predictions from patients the model did not train on

A model is too optimistic on patients it trained on. This is worst in small subgroups.

- With cross-validation, use each patient's prediction from the fold that left the patient out.
- Resampling such as SMOTE (synthetic minority oversampling) belongs inside the training folds only.

In the worked example, the model was trained with SMOTE. Refitting without it moved the overall calibration intercept by +0.94 logits (0.91 to 0.96), in 2018–2022 data. The frail minus pre-frail difference grew by 0.17.

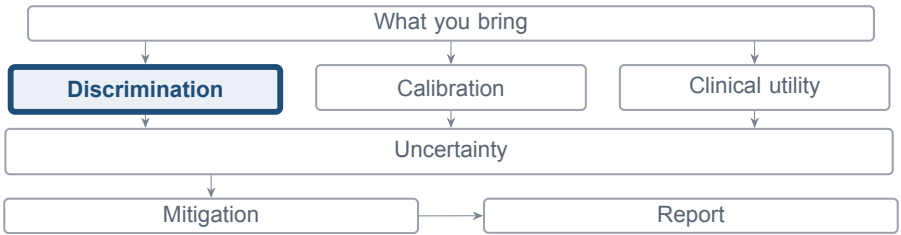
Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Supplementary Table 2.

Construct the audit

```
from isitfair import FairnessAudit

audit = FairnessAudit(
    y_true=y,               # 0/1 outcome
    y_proba=p_oof,          # out-of-fold predicted probability
    sensitive_features={
        "sex": sex,
        "age_group": age_group,
        "reserve": reserve,
        ("sex", "reserve"): None,   # intersection, labels built for you
    },
    threshold=0.10,         # fixed before you look at results
    n_bootstrap=2000,
    random_state=42,        # same seed and data give the same output
)
```

Discrimination



AUROC measures how well the model ranks patients

$$\mathrm{AUROC}_g = \Pr\left(p_i > p_j \mid i,j \in I_g,\ Y_i = 1,\ Y_j = 0\right)$$

In words. Take one patient with the event and one without, from the same subgroup. AUROC is the probability that the model gives the higher score to the patient with the event.

- It needs no threshold.
- It changes little when prevalence changes.

Common mistake. Comparing subgroup AUROCs as if all scores were on one scale. Each AUROC compares patients only inside its own subgroup.

AUPRC must be read against the subgroup's prevalence

AUPRC is the area under the precision–recall curve: positive predictive value against sensitivity, over all thresholds.

In words. Of the patients you would flag, how many have the event, summed over all thresholds.

- A useless model has AUROC 0.5, but AUPRC equal to the prevalence π_g^+.
- So AUPRC 0.376, as in the worked example, is good at 4.70% prevalence. It would be poor at 30%.

Common mistake. Ranking subgroups by AUPRC. This mostly ranks their prevalence.

A threshold turns the score into a decision rule

$$\mathrm{TPR}_g = \frac{TP_g}{TP_g + FN_g}, \quad \mathrm{FPR}_g = \frac{FP_g}{FP_g + TN_g}, \quad \mathrm{PPV}_g = \frac{TP_g}{TP_g + FP_g}, \quad \mathrm{NPV}_g = \frac{TN_g}{TN_g + FN_g}$$

TPR is sensitivity, FPR the false-positive rate, PPV and NPV the positive and negative predictive values.

In words. Sensitivity and FPR are shares of the patients with and without the event. PPV and NPV are shares of the flagged and unflagged patients. PPV and NPV change with prevalence even when the model does not change.

Common mistake. Calling a PPV difference bias when prevalences differ.

Run discrimination

```
disc = audit.discrimination()
disc[disc.subgroup != "GAP"][["attribute", "subgroup", "n_events",
                              "auroc", "auroc_ci_low", "auroc_ci_high"]]
```

One long table. For each attribute: one `Overall` row, one row per subgroup and one `GAP` row per gap statistic.

Keys and size	`attribute, subgroup, shrunk, n, n_events, prevalence`
Metrics	`auroc, auprc, sensitivity, specificity, ppv, npv` each with `_ci_low` and `_ci_high`
Gaps	`gap_metric, gap_value, gap_ci_low, gap_ci_high`
Rules	`n_replicates, n_estimable, proportion_nonestimable, not_estimable, event_floor, below_floor, not_estimable_final`

Four parity gaps compare one rate between subgroups

For two subgroups g and g' at the same threshold t:

$$\Delta_{\mathrm{EOpp}} = \mathrm{TPR}_g - \mathrm{TPR}_{g'} \qquad \text{equal opportunity: sensitivity gap}$$

$$\Delta_{\mathrm{PE}} = \mathrm{FPR}_g - \mathrm{FPR}_{g'} \qquad \text{predictive equality: false-alarm gap}$$

$$\Delta_{\mathrm{SP}} = \frac{TP_g + FP_g}{n_g} - \frac{TP_{g'} + FP_{g'}}{n_{g'}} \qquad \text{statistical parity: flag-rate gap}$$

$$\Delta_{\mathrm{EO}} = \max\left(|\Delta_{\mathrm{EOpp}}|,\ |\Delta_{\mathrm{PE}}|\right) \qquad \text{equalized odds}$$

In words. One rate, subtracted between two groups. With more than two subgroups, the gap is the largest difference over all pairs.

Common mistake. Treating these as properties of the model. They describe the decision rule at one threshold.

Cohen's *h* shows whether a gap is large

$$h(p_g, p_{g'}) = 2\left(\arcsin\sqrt{p_g} - \arcsin\sqrt{p_{g'}}\right)$$

In words. A difference between two proportions, rescaled so that a gap near 0 or 1 counts more than the same gap near 0.5.

- A 2-point sensitivity gap means more at 95% than at 50%.
- Usual reading: 0.2 small, 0.5 medium, 0.8 large.
- Leave *h* blank where it does not apply: equalized odds, calibration intercept and slope, net benefit.

Common mistake. Reading a significant gap as a large gap. In a large cohort almost every gap is significant.

Cohen J. Statistical Power Analysis for the Behavioral Sciences. 2nd ed. Lawrence Erlbaum; 1988.

A gap has one of three readings

1. The subgroups are different populations

The gap reflects case mix, and it is context. Apply this reading to every axis in the same way.

2. The score misleads in one subgroup

Wrong risks, or negative net benefit at the planned threshold. This changes what you do next.

3. There are too few events

Show the point estimate, withhold the interval, and say why.

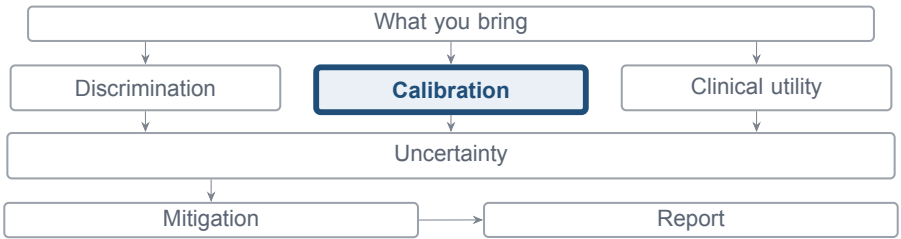
Age showed a threshold gap that sex did not

At $t = 0.10$:

Age	Operations	Infections	AUROC	Sensitivity	Specificity
0–17	367	21	0.807	0.857	**0.462**
18–44	941	34	0.854	0.824	**0.682**
45–64	3,715	147	0.888	0.918	0.666
65–79	4,684	249	0.889	0.912	0.654
80 or older	1,012	53	0.805	0.849	0.577

Equalized-odds gap: 0.010 for sex, 0.220 for age. It comes from specificity: one threshold flags many more uninfected children. Age 0–17 has 21 infections, below the 25-event floor, so its row has no interval.

Calibration



Calibration asks whether a 10% risk means 10%

A model is calibrated in subgroup *g* if, among patients with predicted risk *p*, the observed event rate is also *p*:

$$\mathbb{E}\left[\, Y \,\middle|\, p,\ A = g \,\right] = p$$

In words. Of all patients given 10%, about 1 in 10 should have the event.

- Overestimation: unneeded monitoring, later discharge, avoidable antibiotics.
- Underestimation: patients miss the monitoring they need.

Check it with a curve, with two parameters and with summary scores.

The calibration curve shows where the model is wrong

Plot the observed event rate against the predicted risk, smoothed by LOESS (local regression), with a 95% bootstrap band.

- The diagonal is perfect calibration. Below it, the model overestimates risk.
- Draw each subgroup's curve only over its own range of predictions.
- In a subgroup with few events, draw the curve dashed and without a band.

No single number shows whether the error is small everywhere or large in one part of the risk range. The curve shows it.

The calibration intercept measures the average shift

With log-odds $l_i = \log(p_i/(1-p_i))$, fit one parameter, with the slope fixed at 1:

$$\operatorname{logit} \Pr(Y_i = 1) = \alpha_g + l_i, \qquad i \in I_g$$

In words. One number for “every risk is off by the same amount”.

- $\alpha_g = 0$: right on average.
- $\alpha_g < 0$: risks too high. $\alpha_g > 0$: risks too low.

In the worked example, α = −1.02 overall.

AUROC held between data windows and the intercept moved

	2014–2022	2018–2022
Operations	10,719	6,051
Infections	504	501
Recorded prevalence	4.70%	8.28%
AUROC	0.876	0.855
Calibration intercept α	−1.02	−0.58

Before 2018, an infection needed a present-on-admission code to enter the data, and the code was rarely recorded. Report every result that depends on prevalence together with its data window.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Tables 2 and 3; Supplementary Tables 15 and 16.

The calibration slope measures the spread of the risks

Now fit both terms:

$$\text{logit}\,\Pr(Y_i = 1) = \alpha_g^{(2)} + \beta_g\, l_i, \qquad i \in I_g$$

In words. Are the risks spread too wide or too narrow?

- $\beta_g = 1$: correct spread.
- $\beta_g < 1$: too spread out, typical of overfitting.
- $\beta_g > 1$: too compressed, typical after class rebalancing.

In the worked example, β = 1.57. The intercept $\alpha_g^{(2)}$ of this fit is not the calibration intercept. Do not report it as α.

Read the intercept and slope together

Pattern	Meaning
$\alpha \approx 0, \beta \approx 1$	Calibrated.
$\alpha < 0, \beta \approx 1$	Every risk too high by one constant on the log-odds scale.
$\beta < 1$	Risks too extreme. They need a slope correction.
$\beta > 1$	Risks too compressed. They need a slope correction.

The worked example had both problems: $\alpha = -1.02$ and $\beta = 1.57$. One recalibration for all patients fixed both: α −0.007, β 0.93.

Brier score and ECE summarise calibration in one number

Brier score

$$\mathrm{BS}_g = \frac{1}{n_g} \sum_{i \in I_g} (p_i - Y_i)^2$$

The mean squared difference between risk and outcome. It changes with prevalence.

Expected calibration error (ECE)

$$\mathrm{ECE}_g = \frac{1}{n_g} \sum_{k=1}^{K} n_{g,k} \left| \bar{p}_{g,k} - \bar{y}_{g,k} \right|$$

Sort patients by risk into $K = 10$ equal-size bins. Average the difference between predicted and observed rates. It depends on the bins.

ICI measures the same gap without bins

Integrated calibration index (ICI)

$$\mathrm{ICI}_g = \frac{1}{n_g} \sum_{i \in I_g} \left| f_g(p_i) - p_i \right|$$

The average distance of each patient's risk from the smoothed calibration curve f_g.

ECE and ICI usually agree. If they do not, the bins drive ECE: look at the curve before you report either.

In the worked example: ECE 0.062 and ICI 0.061.

Austin PC, Steyerberg EW. The Integrated Calibration Index (ICI). Stat Med 2019;38:4051-65. doi:10.1002/sim.8281. Values: Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184, Table 3.

Hosmer–Lemeshow cannot prove that a model is calibrated

Sort patients by predicted risk into 10 groups of about equal size. In each group, compare the events the model expected, the sum of the risks, with the events observed.

Null hypothesis: expected = observed in every group

The statistic adds up the differences. The p-value says how surprising the total would be for a calibrated model.

A large p-value is no evidence of calibration

A small p is evidence of miscalibration. A subgroup with 30 events passes almost always, because the test has little power.

Kramer AA, Zimmerman JE. Assessing the calibration of mortality benchmarks in critical care: the Hosmer-Lemeshow test revisited. Crit Care Med 2007;35:2052-6.
doi:10.1097/01.CCM.0000275267.64078.B0

Use g degrees of freedom for out-of-fold predictions

With O_k and E_k the observed and expected events in group k:

$$\chi^2_{\mathrm{HL}} = \sum_{k=1}^{g} \frac{(O_k - E_k)^2}{E_k\,(1 - E_k/n_k)}$$

In words. Each group's miss, squared, divided by how much it could miss by chance. Compare the total with a χ^2 distribution.

- $g - 2$ applies when the model was fitted on the data under test.
- Out-of-fold predictions fit nothing on that data. Use g.

Common mistake. Comparing H–L between subgroups of different size. It grows with n at the same lack of fit.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Supplementary Table 8.

Run calibration

```
cal = audit.calibration()
# calibration_intercept, calibration_slope, brier,
# ece, ici, and GAP rows per attribute

hl = audit.hosmer_lemeshow()              # df = g by default
deciles = audit.hosmer_lemeshow(detail=True)
# per risk group: observed, expected, difference

curves = audit.calibration_curves()       # LOESS curve and band
```

Pass `df_mode="development"` only if the model was fitted on these data. Publish the per-group table together with the H–L statistic.

A calibration gap inherits the event floor of its subgroups

- `max_intercept_gap`: the largest intercept difference between subgroups.
- `max_slope_gap`: the same for the slope.

A gap has no events of its own. If one of its subgroups is below the event floor, the gap has no interval.

In the worked example, the largest reserve gap, 1.62, involves the non-frail subgroup with 6 events. It has no interval. The frail minus pre-frail gap is 0.49 (0.28 to 0.71).

Common mistake. Quoting the largest gap. It usually comes from the smallest subgroup.

The outcome was recorded more completely in frail patients

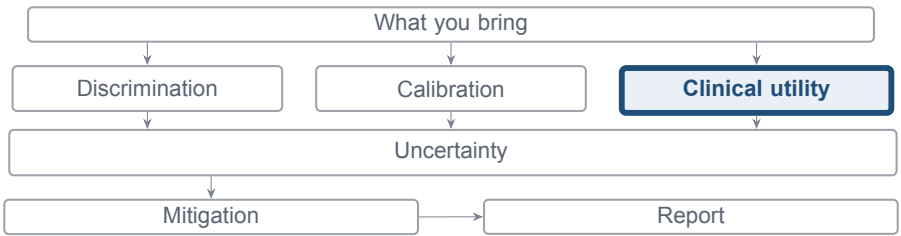
Among patients with an infection code in 2018–2022, the share recorded as an infection after surgery:

Non-frail	15.0% (7.1 to 29.1)
Pre-frail	24.2% (20.2 to 28.6)
Frail	33.5% (30.9 to 36.3)

The present-on-admission field was filled in for 92.5% to 98.0% of patients in each subgroup. Recording may explain part of the reserve gradient, so the gradient describes the model together with the recorded outcome.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Supplementary Table 18.

Clinical utility



A decision curve compares the model with two simple strategies

If I act on this score at threshold *t*, am I better off than if I act on no one, or on everyone?

Draw three curves on one plot:

- The model, at each threshold.
- Treat none: the zero line.
- Treat all: act on everyone, whatever the score.

Treat all must use each subgroup's own prevalence

A whole-cohort reference is the wrong comparison for any subgroup with a different prevalence. Peer review of the companion paper found this error.

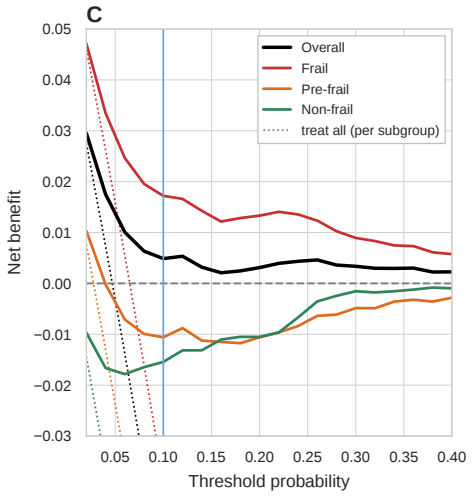
Net benefit counts correct flags and charges for false alarms

$$\mathrm{NB}_g(t) = \frac{TP_g(t)}{n_g} - \frac{FP_g(t)}{n_g} \cdot \frac{t}{1-t}$$

In words. Count the patients correctly flagged. Subtract the false alarms, each weighted by what your threshold says a false alarm costs.

- Units: net true positives per patient. Multiply by 1,000 to report.
- $\mathrm{NB}_g(t) > 0$: better than acting on no one.
- Above the treat-all curve: better than acting on everyone.
- $\mathrm{NB}_g(t) < 0$: acting on the score is worse than ignoring it.

In the worked example, net benefit differed by reserve



Per 1,000 operations at $t = 0.10$, 2014–2022:

Frail	+17.21
Pre-frail	−10.59

Pre-frail in 2018–2022: +3.48 (−5.80 to +13.55).

Non-frail has 6 events, below the 50-event floor for net benefit.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Figure 1C, Table 4 and Supplementary Table 2.

The threshold sets the exchange rate between two harms

Acting at $t = 0.10$ means

$$\frac{t}{1-t} = \frac{0.10}{0.90} = \frac{1}{9}:$$

finding one infection is worth nine unnecessary interventions.

This follows from choosing 0.10

It is not evidence that clinicians accept this exchange. Say which one you have.

Treat the threshold as a reference point, and test it:

- Report net benefit over a plausible range, such as 0.05 to 0.20.
- Draw the curve over a wider range, such as 0.02 to 0.40.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods and Supplementary Table 7.

Standardized net benefit allows comparison between subgroups

Net benefit grows with prevalence. Divide by the subgroup's own prevalence:

$$\mathrm{sNB}_g(t) = \frac{\mathrm{NB}_g(t)}{\pi_g^+}$$

In words. The share of the benefit a perfect model could give in this subgroup. sNB = 1 matches a perfect model. sNB ≤ 0 is no better than acting on no one.

In the worked example at $t = 0.10$, in 2014–2022: frail 0.26, pre-frail −0.38.

Common mistake. Comparing raw net benefit between subgroups. A subgroup with low prevalence has less benefit to gain.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Values from the decision-curve analysis behind Table 4.

The utility gap is the largest disagreement over the range

$$\Delta\mathrm{NB}_{\max} = \max_{t \in [0.02,\, 0.40]} \; \max_{g, g'} \left| \mathrm{NB}_g(t) - \mathrm{NB}_{g'}(t) \right|$$

In words. The largest difference in net benefit between any two subgroups, at any threshold you might plausibly use.

Its interval comes from the same bootstrap replicates as the subgroup values.

The zero crossing shows where the model stops helping

The zero crossing is the threshold at which a subgroup's net benefit becomes zero. isitfair solves it exactly, $t^{\star} = TP/(TP + FP)$, and reports jumps separately.

Worked example, thresholds 0.05 to 0.20:

- Frail: positive at every threshold. No crossing.
- Pre-frail: negative at every threshold. No crossing.
- Age 80 or older: crosses zero 15 times. The last crossing is at 0.135.

A crossing inside the plausible range means the model becomes harmful for that subgroup at a threshold you might use.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Supplementary Table 7.

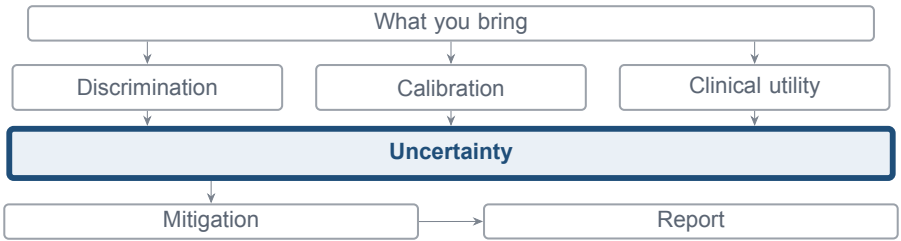
Run clinical utility

```
dca = audit.decision_curve(
    thresholds=[0.05, 0.075, 0.10, 0.125, 0.15, 0.175, 0.20],
    zero_crossing=True,
)
# net_benefit and standardized_net_benefit, each with a CI,
# treat_all_nb, treat_none_nb, zero_crossing, GAP rows

curve = audit.decision_curve()   # default: 20 points, 0.02 to 0.40
```

`treat_all_nb` is already computed per subgroup. Values are per patient: multiply by 1,000 to report.

Uncertainty



The resampling scheme decides what an interval means

Every interval is a 95% percentile interval over 2,000 bootstrap resamples.

`"patient"`, the default

Resample patients within each subgroup. The subgroup size stays fixed. Its **number of events varies**.

`"case_control"`, kept only to reproduce old results

Resample events and non-events separately. Every replicate has the **observed number of events**.

The second scheme is acceptable for AUROC. It makes intervals too narrow for metrics that depend on prevalence: calibration intercept and slope, Brier score, PPV, NPV and net benefit.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods and Supplementary Table 11.

With six events, the two schemes gave very different intervals

Calibration intercept of the non-frail subgroup, 6 events:

Scheme	95% interval	Width (logits)
`case_control`	−2.55 to −2.40	0.15
`patient`	withheld	1.76

The patient interval is more than 10 times wider.

case_control can never draw a replicate without events

So its warning for sparse subgroups always reads 0%. It cannot warn you.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Supplementary Table 11.

A correct bootstrap has three properties

1. One overall interval

The whole cohort is resampled once and reused for every attribute. The overall row is the same in every table.

2. Gaps from the same replicates

All subgroups of an attribute share one set of resampled patients. Each gap is computed from those, which keeps their dependence.

3. Reproducible randomness

Each replicate has its own seed. Results do not depend on order or the number of workers. `random_state=42` gives identical output.

Rule 1: too many failed replicates means no interval

Some replicates cannot produce a value. A resample with no events has no AUROC.

- Failed replicates are stored as missing and counted.
- If more than 10% fail, the result has no interval.
- Every table shows `n_replicates`, `n_estimable` and `proportion_nonestimable`.

The interval uses only the replicates that worked

So it is a little too narrow from the first failed replicate. The 10% limit is a reporting rule.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods and Supplementary Table 10.

Rule 2: each metric needs a minimum number of events

Metric	Minimum events
AUROC	25
Calibration intercept	25
Calibration slope	100
Net benefit	50

In words. Below these counts the metric is mostly noise. The slope needs the most events, because it estimates a gradient.

Common mistake. Setting floors after seeing the results. Fix them before the analysis.

A result needs both rules to get an interval

no interval = below its event floor **or** more than 10% failed replicates

- Below the floor, a result has no interval even if every replicate worked.
- A gap fails if any of its subgroups fails.
- Apply the rules even when they remove your main finding.

Withhold the interval, never the point estimate

Show the point, label it not estimable, and do not interpret it. In figures, use a hollow marker and no interval.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods; Figures 1 and 2.

Ask the audit which results may carry an interval

```
audit = FairnessAudit(
    y, p_oof, groups, threshold=0.10,
    resample_scheme="patient", n_bootstrap=2000, random_state=42,
    nonestimable_threshold=0.10,       # rule 1
    event_floors={                     # rule 2
        "auroc": 25, "calibration_intercept": 25,
        "calibration_slope": 100, "net_benefit": 50,
    },
)
est = audit.estimability()
# metric, n_events, event_floor, below_floor,
# proportion_nonestimable, not_estimable_final, rule
est[est.not_estimable_final]   # results without an interval
```

Save this table next to your results. Later code should read it instead of re-applying the floors.

Correct for multiple testing within each family of tests

Use the Benjamini–Hochberg false discovery rate (BH-FDR) at $q = 0.05$, within each prespecified family. Sort the p-values, find the largest k with $p_{(k)} \leq (k/m)\, q$, and reject the first k.

In words. The more tests in a family, the stricter each test. FDR limits the share of false findings among your findings.

Families in the worked example: the gaps of each axis, a 23-test sex by comorbidity screen, and the Table 1 comparisons. Subgroup estimates are descriptions, so they are not corrected.

Common mistake. Putting every p-value of a paper into one family.

Benjamini Y, Hochberg Y. Controlling the false discovery rate. J R Stat Soc B 1995;57:289-300. doi:10.1111/j.2517-6161.1995.tb02031.x

An interaction test compares an effect between groups

A gap shows what one threshold does. To ask whether a risk factor X acts differently by attribute A, compare two logistic models:

$$M_0: \quad \operatorname{logit} \Pr(Y{=}1) = \beta_0 + \beta_X X + \beta_A A + \beta_C^\top C$$
$$M_1: \quad \operatorname{logit} \Pr(Y{=}1) = \beta_0 + \beta_X X + \beta_A A + \beta_{XA} XA + \beta_C^\top C$$

$$\mathrm{LR} = 2\big(\ell(M_1) - \ell(M_0)\big) \sim \chi^2_{df}$$

In words. Does the extra term improve the fit enough to justify it? The test is for $\beta_{XA} = 0$.

Report the interaction p-value with its effect size

```
from isitfair import interaction_test

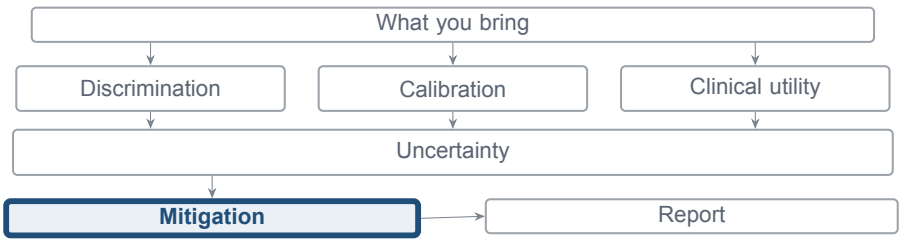
res = interaction_test(
    df, outcome="y", sensitive="sex", covariate="comorbidity",
    adjust_for=("age_years", "asa_class"),   # numeric columns
)
res.p_value, res.delta_r2
```

`delta_r2` is the increase in Nagelkerke R^2 from the interaction term.

In the worked example, sex by physiological reserve gave p = 0.018 and $\Delta R^2 = 0.0008$. The effect is significant and very small.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Results, secondary interaction analyses.

Mitigation


What you bring
Discrimination
Calibration
Clinical utility
Uncertainty
Mitigation
Report

Recalibrate in three steps

Step 0, none. The baseline for every comparison.

Step 1, common. One calibrator for all patients. It removes the shift that affects everyone.

Step 2, group-specific. One calibrator per subgroup. Use it only for the error left after step 1.

In the worked example, step 1 moved α from −1.02 to −0.007 and left a frail minus pre-frail gap of 0.31. Step 2 reduced that gap below 0.01.

Isotonic recalibration changes risk values and keeps their order

Per subgroup g, fit a step function ι that never decreases:

$$\iota_g = \arg\min_{\iota} \sum_{i \in I_g^{\mathrm{cal}}} \left(\iota(p_i) - Y_i\right)^2, \qquad p_i^{\mathrm{iso}} = \iota_g(p_i)$$

In words. Fit a staircase that only goes up. The order of the patients stays the same. Only the values change.

- AUROC is unchanged, except for ties in very small subgroups.
- It can fix curve shapes that a logistic recalibration cannot.

Common mistake. Trusting it in small subgroups. Check the slope after recalibration together with the event count.

Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. ICML 2005:625-32. doi:10.1145/1102351.1102430

Fit the recalibration with cross-fitting

```
ladder = audit.recalibration_ladder(
    by="reserve",
    method="isotonic",
    folds=model_fold_ids,     # the model's own CV folds
)
# one row per step ("none", "common", "group_specific")
# and subgroup: intercept, slope, ECE, ICI, Brier,
# net benefit at t, and the intercept gap
```

Each calibrator is fitted on four folds and applied to the fifth. Every patient gets a recalibrated risk from a calibrator that did not see them.

Common mistake. Drawing new folds when the model was cross-validated. Pass the model's own fold IDs.

Group-specific recalibration collapsed with six events

- Non-frail: 6 events, below every event floor.
- Its isotonic map gave the 1,022 patients only 17 different risk values.
- Slope after recalibration: **0.12** (against 0.93 overall).

The average looks corrected because the map is almost constant.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Table 4 and Supplementary Figure 1.

The worked example used common recalibration only

After step 1, the frail minus pre-frail gap was 0.31 logits.

Differences in how the outcome was recorded could explain up to 0.33 logits in pre-frail patients.

So the authors chose common recalibration. They deferred group-specific recalibration until the outcome is recorded in the same way in every subgroup.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Table 4 and Discussion.

Wasserstein post-processing aligns score distributions

$$Q_{\text{bary}}(u) = \sum_g w_g \, F_g^{-1}(u), \qquad p^W = Q_{\text{bary}}\big(F_g(p)\big)$$

In words. Line up each subgroup's scores by percentile and average them. The order inside each subgroup is kept, so subgroup AUROC does not change.

```
from isitfair import WassersteinPostprocessing

w = audit.mitigate(
    method=WassersteinPostprocessing(constraint="demographic_parity"),
    calibration_data=(y_cal, p_cal, groups_cal),
)
```

Fit it on a separate calibration set. The audited patients stay unseen.

Aligned score distributions make calibration worse

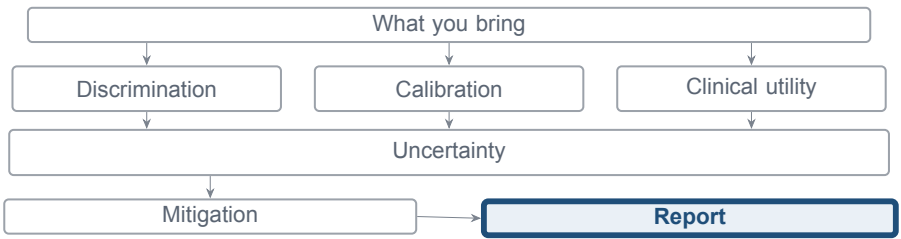
When prevalence differs between subgroups, a score cannot satisfy both calibration and equalized odds.

- Use Wasserstein post-processing as a comparison for parity.
- Expect calibration to get worse.
- If the problem is miscalibration, use recalibration instead.

In the worked example, it was a comparison on the procedure-cluster axis.

Pleiss G, Raghavan M, Wu F, et al. On fairness and calibration. NeurIPS 2017. Example: Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184, Supplementary Table 20.

Reporting


What you bring
Discrimination
Calibration
Clinical utility
Uncertainty
Mitigation
Report

Write the report

```
audit.report(
    out="audit_report.html",
    title="My model: subgroup audit",
    metadata={"Model": "XGBoost v3", "Threshold": "0.10",
              "Predictions": "out-of-fold, 5-fold CV"},
    mitigation_audits=mitigated,   # optional before and after section
    table_one_data=cohort_df,      # optional cohort table
    pdf=True,                      # needs the isitfair[pdf] extra
)
```

One HTML file with all figures inside and no JavaScript. Figures are also saved as 300-DPI PNG files. With a fixed `random_state`, the report is identical on every run.

What the report contains

1. Header and your metadata.
2. Audit summary: patients, events, prevalence, threshold, replicates.
3. Table 1 with effect sizes and BH-FDR.
4. Discrimination: tables, Cohen's *h*, forest plot, ROC curves.
5. Calibration: tables, curves, Hosmer–Lemeshow with per-group counts.
6. Clinical utility: gap table and decision curves.
7. Mitigation before and after, calibration first.
8. Intersection cells: size, events, shrinkage.
9. TRIPOD+AI items covered, and items you must supply.
10. How to read the report.

Small intersection cells are pulled toward their parent subgroup

Cells with fewer than 50 events have their rate shrunk toward the parent subgroup's rate:

$$\tilde{r}_c = (1 - w_c)\, r_c + w_c\, r_0, \qquad w_c = \frac{\tau}{n_c^+ + \tau}$$

In words. The fewer events a cell has, the more its rate is pulled toward the parent. $w_c = 0$ means no shrinkage, $w_c = 1$ full pooling.

- Only rates are shrunk. AUROC and AUPRC are not.
- `cell_diagnostics()` reports w_c for every cell.

Common mistake. Reading a shrunk cell as an estimate of that cell alone.

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Methods and Supplementary Table 12.

The whole audit in code, part 1

```
from isitfair import FairnessAudit

audit = FairnessAudit(
    y_true=y, y_proba=p_oof,
    sensitive_features={"sex": sex, "age_group": age,
                        "reserve": reserve,
                        ("sex", "reserve"): None},
    threshold=0.10, n_bootstrap=2000, random_state=42,
)

results = audit.run_all()
# discrimination, roc_curves, calibration, calibration_curves,
# hosmer_lemeshow, decision_curve, estimability
```

The whole audit in code, part 2

```
for name, table in results.items():
    table.to_csv(f"audit_{name}.csv", index=False)

audit.report(out="audit_report.html",
             title="Subgroup audit")
```

Later scripts read `audit_estimability.csv`. They do not re-apply the event floors.

compare_periods puts two audits side by side

```
from isitfair import compare_periods

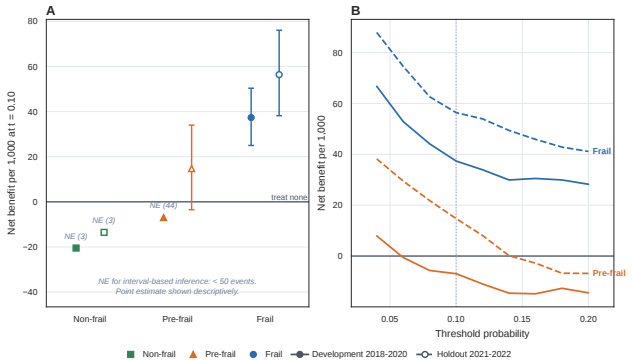
dev  = FairnessAudit(y_dev,  p_dev,  groups_dev,  threshold=0.10)
test = FairnessAudit(y_test, p_test, groups_test, threshold=0.10)

by_period = compare_periods({"2018-2020": dev, "2021-2022": test})
```

isitfair does not refit your model. Your own pipeline refits on the first period, with all preprocessing, feature selection and resampling fitted on that period only.

Common mistake. Reading a difference between periods as model drift. Recording practices also change over time.

Out of period, ranking held and overestimation persisted



Refit on 2018–2020 (3,847 operations, 289 infections). Tested on 2021–2022 (2,204 operations, 212 infections).

AUROC 0.875 then 0.851.
Intercept −0.88 then −0.71.

Pre-frail net benefit per 1,000: −6.94 (44 events, no interval), then +14.72 (−3.49 to +34.00).

Guillen-Ramirez H, et al. Subgroup deployment-readiness audit in perioperative infection prediction. JAMIA 2026 (accepted). doi:10.1093/jamia/ocag184. Table 5 and Figure 4.

Fix these decisions before you look at any result

1. The subgroup axes, and which of them are case-mix axes.
2. The primary axis for your use, and so what counts as a finding.
3. The threshold, and the range you will test around it.
4. The event floors, with a written reason.
5. The test families for FDR correction.
6. That predictions are out-of-fold, and that any recalibration will be cross-fitted.

What to write in a finished audit

- Subgroup values on all three axes, with intervals where the data allow, and “not estimable” where they do not.
- Gaps with effect sizes, read against case mix and prevalence.
- The primary criterion, and what it means for this score at this threshold.
- The mitigation steps you tried, in order, and the ones you rejected, with reasons.
- What is still unresolved, and which data would resolve it.

Do not write that the model is fair. None of these numbers can show that.

Six more public datasets to practise on

The folder `extra_examples/` runs the same audit on six public datasets.

Notebook	Outcome	Test set	Outcome rate
`diabetes130`	Readmission within 30 days	30,530	11.2%
`adult`	Income above 50,000 US dollars	14,653	23.9%
`bank`	Term deposit subscribed	13,564	11.7%
`compas`	New arrest within two years	1,852	45.5%
`german`	Bad credit risk	300	30.0%
`heart_cleveland`	Coronary disease on angiography	91	46.2%

In the two smallest datasets, most results are not estimable: 159 of 208 in `german` and 191 of 208 in `heart_cleveland`.

How to start

```
pip install git+https://github.com/HugoGuillen/isitfair.git
```

Python 3.10 to 3.13.

- **Your own data:** `examples/audit_your_own_data.py`, a template script. The guide is `docs/user_guide/your_own_data.md`.
- **A full example:** `examples/support2_mortality_audit.py` and notebooks on SUPPORT2, a public dataset of 9,105 adults. It shows the software. It does not validate a clinical model.
- **Six more datasets:** `extra_examples/`, with notebooks.
- **The paper:** Guillen-Ramirez et al., JAMIA 2026, doi:10.1093/jamia/ocag184.

Questions are welcome. You can also bring your own model.